

BIMS-LEGAL: Dual-Store Soft Binding for Recovering Prior Legal Advice under Same-Domain Interference

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Abstract

In same-domain legal consultation archives, dense retrieval often surfaces a relevant client question while failing to return the lawyer’s advice turn—an *episode incompleteness* failure that leaves prior recommendations hard to recover under lexical interference from neighbouring matters. BIMS-LEGAL indexes Chinese consultation logs in a dual store: a fast episodic turn index with session identifiers and a slower semantic store consolidated by target- k KMeans. The design target is episode completeness under same-domain interference: Soft O2 binds session siblings by attenuated score inheritance (βs) rather than hard score copy, so a retrieved question can surface its advice turn without dumping an entire session into top- k . Exact replay is a surface-form reference; main channels are paraphrase, follow-up, and advice-recall. On a shared turn-level FlatIP rebuild, Soft O2 raises Answer Hit over dense FlatIP; hard hydration, which copies sibling scores without attenuation, can raise AH further by forcing siblings into top- k . Soft O2-C is evaluated as a conditional cluster-binding mechanism under Mix reconstructions with off-session gold; gated Hybrid is the composed operator when thematic co-membership helps. We report Answer Hit, Episode Completeness, graded nDCG with fixed gold-session IDCG, a four-way failure taxonomy, and Holm-

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adjusted Soft O2 vs FlatIP tests that reuse the same per-query Answer Hit outcomes as the Soft O2 tables.

Keywords: legal information retrieval, consultation memory, soft episode binding, dual-store memory, cluster binding, Chinese legal corpora

1. Introduction

A junior associate may ask a firm assistant: “Last month you advised this client on terminating a fixed-term contract—what remedy did we recommend?” The system returns several near-identical “contract termination” questions, ranks the stored user utterance highly, and never surfaces the partner’s written advice. The subsequent reply may sound fluent yet be grounded in the wrong episode—or in none. In consultation archives, questions and answers share specialized terminology, neighbouring matters are topically adjacent yet legally distinct, and the professionally useful evidence unit is the full consultation *episode*.

Legal informatics has long studied statutory interpretation, case retrieval, and advisory systems Martinez-Gil (2023); Ashley (2017), and recent legal large language models (LLMs) further improve statutory and fact-pattern answering Yue et al. (2023); Huang et al. (2023); Yue et al. (2024). Classical legal information retrieval (IR), however, mainly targets *external* authority—statutes, cases, or community answers—as documents Askari et al. (2024); Louis et al. (2024); Liu et al. (2022); Zhang et al. (2024). Memory-augmented dialogue systems recover evidence from evolving conversation logs Wu et al. (2024); Maharana et al. (2024); Packer et al. (2023), yet firm assistants still need a reliable index of *internal consultation memory*: prior client-specific advice buried in a growing same-domain archive under lexical interference from neighbouring matters.

Under that interference, dense turn retrieval often returns a question turn while its answer sibling remains outside top- k . We term this failure *episode incompleteness*, as distinct from a full session miss. It is worsened when approximate indexes distort turn scores and fragment episode context at moderate archive sizes. Parent-document hydration and session-max aggregation can force siblings into the result list by hard score copy, but they also erase rank competition among unrelated candidates. Soft O2 therefore completes missing siblings by attenuated score inheritance rather than hard score copy.

32 This paper presents BIMS-LEGAL, a dual-store system for Chinese legal
 33 consultation logs. A fast episodic pathway indexes timestamped turn em-
 34 beddings with speaker role and session identifier *sid*; a slower semantic path-
 35 way consolidates turns with target- k KMeans. We adopt a dual-pathway
 36 layout—fast episode index; slower semantic consolidation—as an architec-
 37 tural analogy to Complementary Learning Systems McClelland et al. (1995);
 38 Kumaran et al. (2016), without claiming a neuroscientific equivalence. On
 39 the episodic pathway, Soft O2 performs session soft binding: when a turn
 40 is retrieved with score s , siblings sharing *sid* inherit $\max(s_\nu, \beta \cdot s)$, so an
 41 answer turn can surface under question-hit / answer-miss while attenuated
 42 scores keep candidate competition intact. On the semantic pathway, Soft O2-
 43 C applies the same rule within KMeans clusters when gold evidence is not
 44 co-session with the query; gated Hybrid is intended to trigger cluster binding
 45 only on direct dense hits. Exact FlatIP indexing (O1) restores faithful turn
 46 scores when product quantization collapses Answer Hit; bulk-load consolida-
 47 tion (O3) makes large shared-store construction tractable without changing
 48 retrieval semantics. We evaluate Soft O2 primarily against FlatIP under
 49 episode incompleteness, treat hard hydration as an unrestricted score-copy
 50 upper bound, and examine Soft O2-C under Mix reconstructions as a condi-
 51 tional mechanism test.

52 The main contributions are as follows. First, we formulate consultation-
 53 memory retrieval under same-domain interference and identify episode in-
 54 completeness as a recurrent failure mode of dense turn retrieval. Second,
 55 we propose Soft O2 as a turn-level soft-binding operator: a gap-ratio score-
 56 competition analysis identifies a feasible high- β band $\rho < \beta < 1$, and an Answer
 57 Hit sensitivity sweep over $\{0.5, 0.7, 0.9, 0.95, 0.98, 1.0\}$ confirms a high-AH
 58 plateau from which we freeze default $\beta=0.98$ (Table A.4), with O1/O3 as
 59 implementation operators for shared-store evaluation. Third, we evaluate
 60 Soft O2 on CAIL multi-turn and rebuilt LegalEp corpora against FlatIP,
 61 hard hydration, and shuffled-*sid* under a shared FlatIP rebuild, with BM25-
 62 turn, RRF, and cross-encoder controls as secondary comparisons, and report
 63 Soft O2-C Mix contrasts as conditional mechanism evidence with graded
 64 nDCG, a four-way failure taxonomy, and Holm-adjusted Soft O2 vs FlatIP
 65 tests.

66 Section 2 reviews related work. Section 3 presents BIMS-LEGAL and
 67 the Soft O2 attenuation analysis. Section 4 details corpora, metrics, and
 68 protocols. Section 5 reports results. Section 6 discusses implications and
 69 limitations, and Section 7 concludes.

2. Related Work

2.1. Legal IR and long-horizon dialogue memory

Legal QA and retrieval systems typically ground generation in statutes, cases, or published advice Louis et al. (2024); Askari et al. (2024); Martinez-Gil (2023); Ashley (2017). Conversational legal case retrieval and event-aware similar-case ranking further show how interaction context and structured legal knowledge shape IR effectiveness Liu et al. (2022); Zhang et al. (2024). Those pipelines optimize access to external authority. Consultation memory retrieval targets a different evidence source: *prior consultation episodes* inside an agent’s own archive. Confusing two clients’ histories can produce fluent yet professionally inappropriate responses even when statute retrieval is correct. Domain LLMs such as DISC-LawLLM, Lawyer LLaMA, and LawLLM improve statutory and advisory generation Yue et al. (2023); Huang et al. (2023); Yue et al. (2024), but they still need a reliable memory index when prior advice must be recovered from a growing same-domain log.

Legal consultation archives compound that difficulty: high lexical overlap among topically adjacent matters, with the professionally useful evidence unit being the full question–answer episode.

2.2. Dual-store memory and episodic retrieval

Dual-store designs distinguish fast episodic encoding from slower semantic integration McClelland et al. (1995); Kumaran et al. (2016); Tulving et al. (1972); Atkinson and Shiffrin (1971). Memory architectures for dialogue agents extend context via hierarchical paging, summary banks, graphs, and neuro-symbolic retrieval Packer et al. (2023); Zhong et al. (2024); Rasmussen et al. (2025); Gutiérrez et al. (2024); Modarressi et al. (2023); Wang et al. (2024); Lewis et al. (2020); Wu et al. (2025), but most treat retrieved units as flat passages rather than *multi-turn consultation episodes* whose answer turn may rank below an unrelated question turn sharing legal vocabulary. LongMemEval and LoCoMo evaluate long-horizon recall and QA over personal conversation logs Wu et al. (2024); Maharana et al. (2024), while broader memory-agent surveys emphasize persistence beyond the context window Wu et al. (2025); Tan et al. (2025). Recent conversational RAG work stresses reuse of historical search evidence across turns Mo et al. (2026). BIMS-LEGAL indexes episodic turns with *sid* (Soft O2) and thematic KMeans clusters (Soft O2-C) when query and gold evidence occupy different sessions but share thematic structure. Under dense same-domain

interference, a question turn can occupy top- k while its answer sibling is absent. Parent-document hydration and session-max aggregation recover that sibling by hard score copy; Soft O2 instead injects attenuated inherited scores so episode completion does not erase rank competition among unrelated candidates.

2.3. Session-aware retrieval

Parent-document hydration, session aggregation, and joint session indexing are standard session-aware mechanisms for attaching sibling context to a dense hit. Soft episode binding (Soft O2) inherits $\beta \cdot s$ for siblings and keeps ranking competition among candidates. Parent-document and hierarchical retrieval similarly attach child evidence to a parent unit Callan et al. (1994); Dai et al. (2019); Zaheer et al. (2017); Karpukhin et al. (2020); Soft O2 differs by using attenuated turn-level score propagation rather than hard hydration or atomic session documents. Lexical BM25 Robertson and Zaragoza (2009), dense passage retrieval Karpukhin et al. (2020), dense-sparse Reciprocal Rank Fusion Cormack et al. (2009), and cross-encoder reranking Nogueira et al. (2019); Chen et al. (2024) provide complementary controls outside session membership. Exact flat indexes and product-quantized ANN backends Johnson et al. (2019) further affect whether sibling binding is applied over faithful turn scores.

Contrast with joint-episode indexing. An alternative way to keep episode context is to concatenate full question-answer pairs into monolithic episode documents (joint indexing). Joint indexing can prevent episode incompleteness by construction, but it changes the evidence unit from individual turns to merged documents. That coarser unit weakens fine-grained turn matching—for example, isolating a specific advice sentence—and raises index memory and update cost when new turns arrive in an online consultation log. Soft O2 instead keeps turn-level representations and candidate competition, recovering missing siblings on demand via attenuated score inheritance (βs) without replacing the underlying turn index. Table 1 summarizes the session-structure contrasts used in the main grids.

*On the reported corpora, session-max rankings coincide with hard parent hydration under full-sibling expansion; main tables report a single hard-expansion baseline. Soft O2 denotes session soft binding; Soft O2-C denotes cluster soft binding on the KMeans semantic pathway.

Table 1: Session-aware mechanisms compared in this work.

Mechanism	Unit	Expansion	Type	Rank
Parent hydration	Session	Insert siblings	Hard	No
Session-max	Session	Session max	Hard	Same*
Joint episode index	Session doc	Atomic	N/A	N/A
Soft O2	Turn+ <i>sid</i>	Soft βs	Soft	Yes
Soft O2-C	Turn+cluster	Soft $\beta_c s$	Soft	Yes

3. Method: BIMS-LEGAL

3.1. Architecture overview

Figure 1 summarizes the pipeline. BIMS-LEGAL maintains complementary episodic and semantic stores over the same consultation archive.

Definition 1 (Episodic and semantic elements). *An episodic turn (episodic element) is $e_i = (\mathbf{v}_i, t_i, c_i, \phi_i, \text{sid}_i)$, where $\mathbf{v}_i \in \mathbb{R}^d$ is an embedding, t_i a timestamp, c_i conversational context (including speaker role), ϕ_i surface features, and sid_i the consultation session identifier. We reserve session turns for the weaker notion of turns that share only *sid* under a grouping key, without requiring role, timestamp, or other episodic metadata to participate in scoring. Soft O2 binds over session siblings $\text{Sib}(t) = \{t' : \text{sid}_{t'} = \text{sid}_t\} \setminus \{t\}$, but the indexed objects themselves remain episodic turns. A semantic element is $s_j = (\mathbf{c}_j, C_j, \kappa_j, \tau_j)$ with centroid $\mathbf{c}_j$, member set C_j , label κ_j , and formation statistics τ_j . Write $\text{Clus}(t) = \{t' : \text{cid}_{t'} = \text{cid}_t\} \setminus \{t\}$ for semantic-cluster siblings.*

Definition 2 (Dual-store retrieval map). *Given query q with embedding $\mathbf{v}_q$, base episodic retrieval returns a scored candidate set $\mathcal{R}_0(q) = \{(t, s_t)\}$ under Eq. (1). Soft O2 produces $\mathcal{R}_{\text{O2}}(q) = \text{SoftBind}(\mathcal{R}_0(q); \text{Sib}, \beta)$ (Section 3.3). Soft O2-C produces $\mathcal{R}_{\text{O2C}}(q) = \text{SoftBind}(\mathcal{R}_0(q); \text{Clus}, \beta_c)$ (Section 3.5). Gated Hybrid composes session binding then cross-session cluster binding (Algorithm 2). The returned list is the top- k of the bound set under the rewritten scores.*

3.2. Base scoring and implementation operators (O1, O3)

For query q with embedding $\mathbf{v}_q$, episodic base retrieval scores a memory turn m by

$$\text{score}(m) = \alpha S(\mathbf{v}_q, \mathbf{v}_m) + (1 - \alpha) R(t_m), \quad (1)$$

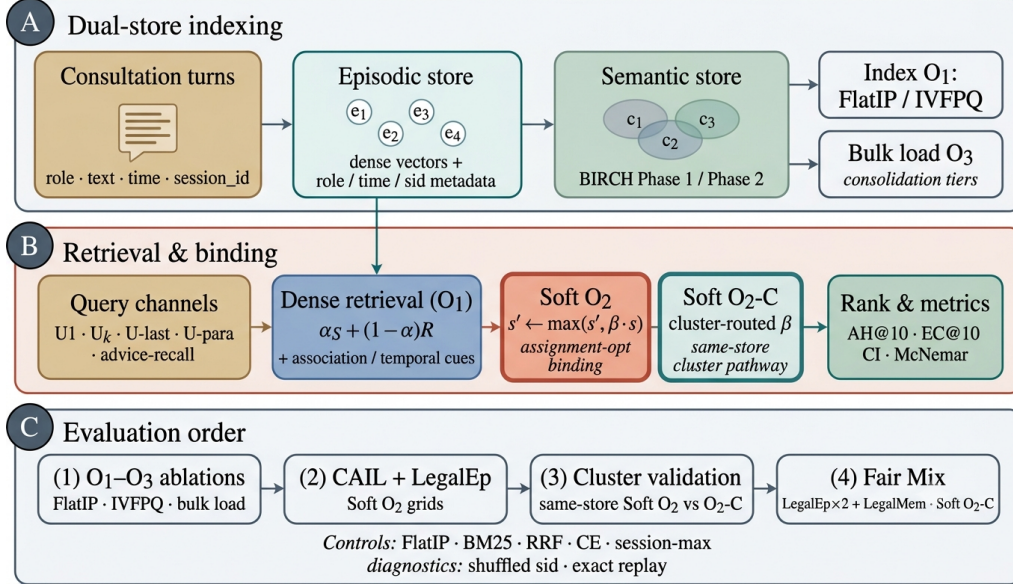


Figure 1: BIMS-LEGAL dual-store pipeline. (A) Episodic turn indexing and KMeans semantic consolidation (O3 bulk-load). (B) Query-time FlatIP retrieval with Soft O2 session binding and Soft O2-C cluster binding; O1 and O3 are implementation operators. (C) Evaluation order: shared-store ablations, Soft O2 grids, and Mix Soft O2-C.

166 where S is cosine similarity on ℓ_2 -normalized embeddings and

$$R(t_m) = \max\left(0, 1 - \frac{\Delta_t}{30 \text{ days}}\right), \quad (2)$$

167 with Δ_t the elapsed time between query time and the turn timestamp. Unless
 168 stated otherwise, $\alpha=0.7$. Eq. (1) is the ranking score used throughout the
 169 legal grids; Soft O2 /Soft O2-C rewrite sibling scores after this fusion step
 170 and do not replace Eq. (1).

171 *O1: Exact FlatIP indexing..* Let $\hat{S}$ denote the similarity induced by an ANN
 172 backend. At hundreds to a few thousand turns, FAISS IVFPQ is often under-
 173 trained, so $\|\hat{S} - S\|$ is large enough to reorder turns and break Soft O2
 174 inheritance. O1 forces $\hat{S} = S$ via exact FlatIP (inner-product search on
 175 ℓ_2 -normalized embeddings). Quantization error depresses Answer Hit (Ta-
 176 ble A.3: IVFPQ AH 0.330 at $M=1600$ on LegalEp-DISC exact). O1 is an
 177 engineering prerequisite for Soft O2 evaluation, not a standalone retrieval
 178 algorithm.

179 *O3: Bulk-load consolidation..* During bulk construction, O3 replaces the per-
 180 insert update ($e_i \mapsto \text{KMeansConsolidate}(e_i); \text{Flush}$) by a deferred operator
 181 $\text{Finalize} = \text{KMeansConsolidate}(\{e_i\}_{i=1}^N) \circ \text{Flush}$, executed once after all episodic
 182 writes. O3 does not change retrieval semantics; it makes $M \geq 400$ shared-
 183 store experiments runnable (≈ 11 min wall-clock for $M=400$; amortized $\approx$
 184 0.825 s/turn over ~ 800 turns). Quality-latency curves to $M=6400$ appear
 185 in Section 5.6.

186 3.3. Soft O2: session soft binding

187 *Motivation..* Under same-domain interference, a retrieved question turn can
 188 score highly while its answer sibling remains outside top- k —episode incom-
 189 pleteness compounded by IVFPQ score collapse and context fragmentation.
 190 Soft O2 completes episodes without hard score copy.

191 **Definition 3** (Soft binding operator). *Given a scored set $\mathcal{R} = \{(t, s_t)\}$, a*
 192 *sibling relation $\mathcal{N}(\cdot)$, and attenuation $\beta \in (0, 1]$, define*

$$\text{SoftBind}(\mathcal{R}; \mathcal{N}, \beta) : \quad s_{t'} \leftarrow \max(s_{t'}, \beta \cdot s_t) \quad \text{for all } t \in \mathcal{R}, t' \in \mathcal{N}(t). \quad (3)$$

193 *Soft O2 is $\text{SoftBind}(\cdot; \text{Sib}, \beta)$ with default $\beta=0.98$. Hard parent hydration is*
 194 *the special case $\beta=1$ with forced insertion of all siblings; session-max replaces*
 195 *each member score by $\max_{t \in \text{sid}} s_t$.*

196 The released Soft O2 pipeline expands session siblings on raw dense scores,
 197 optionally applies exact-match boost and sparse augment, performs session-
 198 first reranking, fuses semantic and recency scores (Eq. (1)), then runs a final
 199 completeness pass before top- k truncation. Algorithm 1 records that full
 200 stage order; β acts on the score field present at each expansion stage in the
 201 released memory manager.

202 3.4. Soft O2 attenuation and score competition

203 The attenuation β controls how strongly a retrieved turn promotes its
 204 session siblings. Availability of the injected sibling rises with β , while fidelity
 205 of the direct hit falls as $\beta \rightarrow 1$. Table 2 summarizes the empirical gap-
 206 ratio distribution $\rho = s_d/s_h$ on LegalEp stores under an exact-channel query
 207 proxy: median ρ lies well below one, so a high- β point near 0.98 covers most
 208 measured distractors while remaining strictly below hard hydration ($\beta=1$).
 209 A three-candidate score-competition model yields the feasible band $\rho < \beta < 1$;
 210 we then confirm the operating region with an Answer Hit sensitivity sweep
 211 rather than treating a closed-form hinge as a selector (a zero-margin hinge
 212 on $\beta \in (0, 1]$ is uninformative because its rank term vanishes for all $\beta < 1$).

Algorithm 1 Soft O2 session soft binding (implementation order)

Require: Query q , dense/associative hits $\mathcal{R}_0$, top- k , attenuation β

- 1: $\mathcal{R} \leftarrow \text{ExpandSiblings}(\mathcal{R}_0; \beta)$ $\triangleright$ raw-score Soft O2
 - 2: $\mathcal{R} \leftarrow \text{ExactBoost/SparseAugment}(q, \mathcal{R})$ $\triangleright$ optional
 - 3: $\mathcal{R} \leftarrow \text{SessionFirstRerank}(\mathcal{R}, k)$
 - 4: fuse each $r \in \mathcal{R}$ with Eq. (1)
 - 5: $\mathcal{R} \leftarrow \text{EnsureSessionCompleteness}(\mathcal{R}; \beta)$ $\triangleright$ final Soft O2 pass
 - 6: **return** top- k of $\mathcal{R}$ by descending fused score
-

Table 2: Gap-ratio statistics on LegalEp $M=3000$ stores (exact-channel query proxy; strongest measured non-session distractor). The feasible band $\rho < \beta < 1$ motivates the high- β operating region later confirmed by the AH grid search in Table A.4.

Corpus	ρ_{50}	ρ_{95}	Cover@0.90	Default β
LegalEp-DISC	0.746	0.854	0.981	0.98
LegalEp-Lawyer	0.662	0.815	0.998	0.98

213 **Definition 4** (Three-candidate score-competition model). *For a query whose*
214 *dense retrieval directly hits a gold turn h with fused score $s_h > 0$, let a be*
215 *the gold answer sibling and let d be the strongest non-session distractor in a*
216 *wide candidate pool, with scores s_a and s_d . Soft O2 rewrites*

$$s'_h = s_h, \quad s'_a = \max(s_a, \beta s_h), \quad s'_d = s_d, \quad (4)$$

217 *and defines the gap ratio $\rho = s_d/s_h$. Because d is the strongest measured*
218 *distractor, conditions based on ρ are conservative sufficient conditions for*
219 *beating that distractor, not exact top- k boundary conditions for every com-*
220 *petitor.*

221 The binding analysis is most relevant in the soft-injection regime $s_a \leq \beta s_h$,
222 where $s'_a = \beta s_h$.

223 **Proposition 1** (Feasible attenuation band). *Under Definition 4, assume*
224 *the soft-injection case $s_a \leq \beta s_h$. Then (i) availability against the measured*
225 *distractor requires $\beta > \rho$; (ii) rank preservation of the direct hit over the*
226 *injected sibling requires $\beta < 1$, and more generally $s'_a < s_h$; (iii) when $\rho < 1$,*
227 *any $\beta \in (\rho, 1)$ satisfies both conditions in the active injection case. If $s_a > s_h$*
228 *already, rank preservation depends on the observed sibling score rather than*
229 *on β alone.*

230 *Proof.* If $s_a \leq \beta s_h$, then $s'_a = \beta s_h$. Availability follows from $\beta s_h > s_d$, i.e.
 231 $\beta > \rho$. Rank preservation requires $s'_a < s_h$, which reduces to $\beta < 1$ when
 232 $s_h > 0$. $\square$

233 *Soft-rank visualization and AH sensitivity sweep.* As a theoretical diagnostic
 234 we also plot the soft pairwise surrogate

$$\mathcal{L}_{\text{sr}}(\beta; \rho) = -\log \sigma(\tau(\beta - \rho)) - \gamma \log \sigma(\tau(1 - \beta)), \quad (5)$$

235 with logistic σ , temperature τ , and rank weight γ . $\mathcal{L}_{\text{sr}}$ visualizes the availability–
 236 ranking trade-off under the measured ρ pool; it is not the selector. Default
 237 $\beta=0.98$ is frozen after an Answer Hit sensitivity sweep over $\{0.5, 0.7, 0.9, 0.95, 0.98, 1.0\}$
 238 on the CAIL Uk-followup and LegalEp exact/paraphrase evaluation channels
 239 (Table A.4). The high-AH plateau $\{0.9, 0.95, 0.98\}$ coincides with $\rho < \beta < 1$;
 240 $\beta=1$ is not better and $\beta=0.5$ collapses binding. The sweep uses the same eval-
 241 uation query pools as the Soft O2 grids—*not* a held-out validation split—and
 242 comes from LegalMem-MT β -campaign stores. Absolute AH in Table A.4 is
 243 therefore not comparable to the unified FlatIP rebuild (Tables 5–A.7), which
 244 freezes $\beta=0.98$ after selection; grid provenance (campaign ids, pool sizes,
 245 overlap flag) ships with the manuscript package. Advice-recall inherits the
 246 same frozen default without a separate gap-ratio fit.

247 Fair controls include hard parent hydration, session-max aggregation, and
 248 shuffled *sid*. On two-turn LegalEp episodes, hard hydration and session-max
 249 coincide under full-sibling expansion, so tables report a single hard-expansion
 250 baseline. Concatenating a full question–answer episode into one retrieval unit
 251 (BM25-joint or dense joint-episode documents) changes the evidence unit
 252 from turns to documents; we treat that design as an alternative indexing
 253 choice rather than a Soft O2 turn-binding baseline (Section 4).

254 3.5. Semantic consolidation and Soft O2-C

255 Semantic consolidation uses target- k KMeans over turn embeddings, pro-
 256 ducing cluster ids consumed by Soft O2-C. Other agglomerative helpers that
 257 may appear in the repository are unused on the Soft O2-C evaluation path.

258 **Definition 5** (KMeans semantic consolidation). *Given embeddings $\{e_i\}$ and*
 259 *target cluster count k , O3 runs $(\text{labels}, \text{centers}) \leftarrow \text{KMeans}(\{e_i\}; k)$ once at*
 260 *finalize time and assigns each turn t a cluster id cid_t , inducing $\text{Clus}(t)$ in Def-*
 261 *inition 1. Empty clusters are dropped and labels are remapped to a contiguous*
 262 *range.*

Algorithm 2 Gated Hybrid Soft O2 + Soft O2-C

Require: Base hits $\mathcal{R}_0$, budgets (k, B_c) , attenuations (β, β_c)

```
1:  $\mathcal{R} \leftarrow \text{SoftBind}(\mathcal{R}_0; \text{Sib}, \beta)$  ▷ session pathway first
2:  $\mathcal{S}_{\text{cov}} \leftarrow \{sid_t : t \in \text{top-}k(\mathcal{R})\}$ 
3:  $\mathcal{T}_{\text{trig}} \leftarrow \{t \in \mathcal{R}_0\}$  ▷ direct dense hits only
4: for each trigger  $t \in \mathcal{T}_{\text{trig}}$  do
5:    $s_{\text{max}} \leftarrow \max\{s_u : u \in \mathcal{R}, cid_u = cid_t\}$ 
6:   for each  $t' \in \text{Clus}(t)$  with  $sid_{t'} \notin \mathcal{S}_{\text{cov}}$ , up to budget  $B_c$  do
7:      $s_{t'} \leftarrow \max(s_{t'}, \beta_c \cdot s_{\text{max}})$ 
8:   end for
9: end for
10: return top- $k$  of  $\mathcal{R}$ 
```

263 **Definition 6** (Soft O2-C). *Soft O2-C is $\text{SoftBind}(\cdot; \text{Clus}, \beta_c)$ with default*
264 *$\beta_c=0.95 < \beta$, reflecting noisier thematic neighbourhoods than session mem-*
265 *bership. A per-cluster sibling budget B_c caps $|\text{Clus}(t) \cap \text{Injected}|$ so large*
266 *thematic clusters cannot flood top- k .*

267 Gated Hybrid (Algorithm 2) injects only *cross-session* cluster siblings
268 relative to sessions already covered by dense /Soft O2 hits, and triggers
269 cluster binding only on direct dense hits. This prevents thematic confusers
270 from displacing same-session siblings already recovered by Soft O2. Soft O2
271 applies when query and gold share *sid*. Soft O2-C is the cluster-pathway
272 operator when gold evidence lies in another session of the same thematic
273 cluster.

274 4. Corpora and Evaluation Protocol

275 Figure 2 sketches the shared-corpus design used throughout the legal eval-
276 uation. Gold needles and same-domain distractors occupy one store. Each
277 query probes that shared archive under fixed interference conditions. Pri-
278 vately rebuilt haystacks are avoided so that system comparisons isolate the
279 retrieval operator.

280 4.1. Datasets and query channels

281 Multi-turn authenticity is evaluated on CAIL2024 consultation dialogues.
282 Gold needles are authentic preliminary and final dialogues, with conversation
283 turns completed by label turns where applicable. Query channels cover the

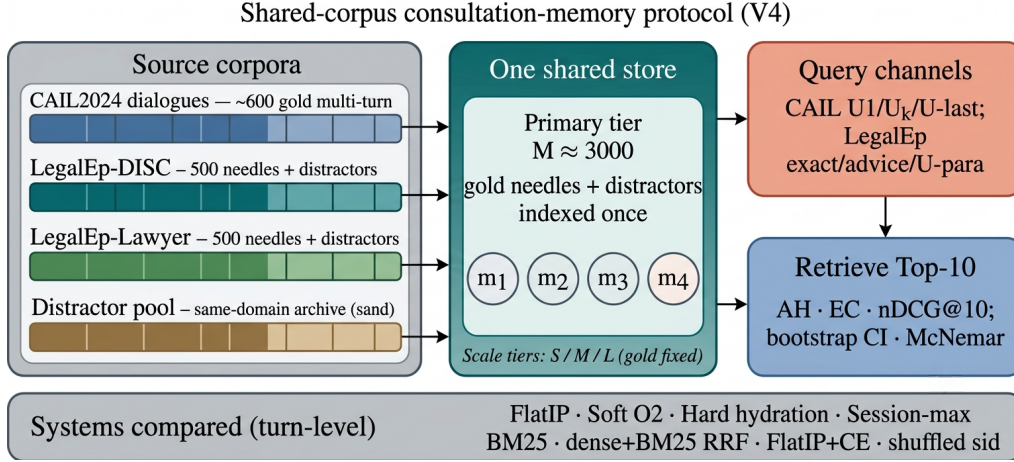


Figure 2: Shared-corpus protocol. Gold needles and same-domain distractors share one store; queries probe multiple channels with turn-level systems and significance tests. Exact replay is diagnostic; paraphrase, follow-up, and advice-recall are primary channels.

284 first user turn ($U1$), a later in-dialogue user turn ($U_k\text{-followup}$), the last user
 285 turn ($U\text{-last}$), and constrained paraphrases ($U\text{-para}$). Evidence defaults to
 286 assistant turns in the target session.

287 LegalEp-DISC and LegalEp-Lawyer rebuild public DISC-Law-SFT and
 288 Lawyer-LLaMA subsets into consultation-episode corpora. Each episode is
 289 a natural (question, answer) pair with session identifier *sid*. Reconstruction
 290 applies length bounds and legal-content filters, excludes exam prompts, re-
 291 moves near-duplicates, separates needles from distractors under a fixed seed,
 292 and retains an audit subsample. After filtering, each LegalEp corpus retains
 293 3500 passed sessions (500 needles, 2500 distractors at the $M=3000$ primary
 294 tier).

295 *Query-channel policy.* Exact replay uses the stored user question and is
 296 reported only as a *diagnostic* upper bound on surface-form recovery. Primary
 297 claims use paraphrase, $U_k\text{-followup}$ / $U\text{-last}$, and advice-recall queries whose
 298 surface form is not identical to the stored question text; paraphrase and
 299 advice-recall templates are generated under a fixed seed and audited on a
 300 subsample. LegalMem-MT reuses CAIL multi-turn gold under the same store
 301 for same-session Soft O2 vs Soft O2-C contrasts and for Mix reconstructions.

302 Archive size is varied with gold needles held fixed. Unless stated otherwise,
 303 Soft O2 transfer tables use the $M \approx 3000$ tier; O1–O2 ablations use $M=400$,

304 $S=300$.

305 4.2. Metrics and baselines

306 Primary metrics are Answer Hit@ k (AH), Episode Completeness@ k (EC),
307 and normalized Discounted Cumulative Gain@ k (nDCG) at $k=10$. AH is
308 binary: whether a gold answer turn appears in the top- k . EC is the mean
309 coverage of gold turns within the target session. Session Hit@ k is reported
310 only for diagnosis. For nDCG we use fixed graded relevance over the full
311 gold session: answer turns $rel=1.0$, other same-session turns $rel=0.5$, all
312 other turns $rel=0.0$. The ideal denominator (IDCG) is computed from all
313 gold-relevant turns in the target session and truncated at k , so every system
314 on a query shares the same IDCG. Answer Hit and Episode Completeness
315 are the main Soft O2 outcomes; nDCG is reported with the fixed gold-session
316 IDCG so that systems remain comparable on each query. Hard expansion
317 can raise nDCG when copied siblings occupy early ranks, so AH and EC are
318 read together with nDCG.

319 The primary hypothesis family is Soft O2 vs FlatIP on the nine CAIL and
320 LegalEp transfer channels. AH, EC, nDCG, failure rates, and Table A.2 (Mc-
321 Nemar mid- p with Holm adjustment) share one FlatIP rebuild and the same
322 per-query Answer Hit outcomes; each Δ AH in Table A.2 equals Soft O2–FlatIP
323 in the Soft O2 tables up to ordinary rounding. Appendix Soft O2 grids re-
324 port AH/EC/nDCG point estimates from that rebuild without separate CI
325 columns. Hard hydration, CE, and Soft O2-C Mix contrasts are secondary.

326 Turn-level dense FlatIP (O1) is the base dense retriever. Soft O2 applies
327 soft sibling score inheritance on the episodic pathway; Soft O2-C applies the
328 same rule within KMeans clusters. Hard parent hydration expands a hit
329 to all siblings by copying the trigger score. Lexical controls include BM25
330 over turns and dense+BM25 Reciprocal Rank Fusion (RRF) under the same
331 turn-level evidence unit. BM25-joint and dense joint-episode indexes retrieve
332 pre-concatenated episode documents and therefore answer a coarser retrieval
333 problem; they are discussed as alternative designs in Section 6.3 rather than
334 Soft O2 turn-binding baselines. A cross-encoder (CE) control reranks the
335 top-30 FlatIP candidates with BAAI/bge-reranker-v2-m3. Shuffled sid , a
336 β sweep, and IVFPQ provide membership, attenuation, and quantization
337 controls.

338 4.3. Fair Mix protocol for Soft O2-C

339 On fully same-session LegalMem grids Soft O2 already recovers gold an-
340 swers that share *sid*, leaving Soft O2-C little exclusive headroom and expos-
341 ing it to thematic confusers. To study off-session recovery we rebuild the
342 *same* LegalEp and LegalMem archives under a Mix protocol: 70% of gold
343 episodes are split into query-side S_a and evidence-side S_b with distinct ses-
344 sion identifiers, while 30% remain intact. Distractors are unchanged. FlatIP,
345 Soft O2, Soft O2-C, and gated Hybrid share unrestricted dense retrieval on
346 this store; we report overall AH and stratum AH on cross-session vs same-
347 session queries.

348 *Cluster co-membership and purity under Mix..* After KMeans consolidation,
349 each cross-session S_a/S_b pair is assigned a shared cluster identifier. This
350 construction uses split metadata available only when building the evaluation
351 store; it does not inject labels at query time. The intent is to test Soft O2-
352 C when thematic co-membership is present. Cluster purity still matters: a
353 shared *cid* that also contains many thematically adjacent distractors narrows
354 Soft O2-C’s effective margin, which is why Hybrid gates cluster injection to
355 direct dense hits and caps the per-cluster budget B_c . If recovery still fails
356 under guaranteed S_a/S_b co-membership, the failure is attributable to binding
357 rather than accidental cluster separation. Natural same-cluster rates without
358 this construction are lower (Section 6.3). Soft O2’s session membership rule
359 is unchanged. LegalEp Mix uses advice-recall queries; LegalMem Mix uses
360 mid-split Uk-followup queries. LongMemEval /LoCoMo numbers from prior
361 runs on this codebase (QA correctness 70.7% / 68.2%) supply long-horizon
362 memory context only Wu et al. (2024); Maharana et al. (2024).

363 5. Experiments

364 Unless noted, all dense retrieval grids use the multilingual encoder **nomic-**
365 **embed-text-v2-moe** Nussbaum et al. (2025) (768-d) as a single shared em-
366 bedding backend, with $k=10$ and seed 42 on the main $M\approx 3000$ stores. The
367 encoder is held fixed so Soft O2 vs FlatIP contrasts isolate binding rather
368 than representation changes; sensitivity to alternative Chinese or multilingual
369 encoders (e.g., **bge-large-zh-v1.5**) is discussed in Section 6.3. A five-seed
370 check on a smaller shared store ($M=400$) is reported later. Primary contrasts
371 appear in Figures 3–5; full numeric grids are in the appendix. Each main
372 table maps to a released JSON artifact in the accompanying repository.

373 5.1. Implementation operators and Soft O2 ablation

374 Table 3 reports the shared-store ablation on the DISC-Law and Lawyer-
 375 LLaMA corpora ($M=400$, $S=300$). IVFPQ baseline Answer Hit is 0.540 on
 376 DISC and 0.397 on Lawyer. O1 alone yields a small gain on DISC (0.567)
 377 and a near-tie on Lawyer (0.397). Exact indexing is therefore a necessary
 378 prerequisite, but it is insufficient alone. O1+Soft O2 raises Answer Hit to
 379 0.780 (DISC) and 0.680 (Lawyer), gains of +0.240 and +0.283 over baseline,
 380 with Episode Completeness 0.888 and 0.840. O3 is enabled for all configura-
 381 tions in this ladder as a construction prerequisite; it does not alter retrieval
 382 scores.

Table 3: O1+Soft O2 ablation on the shared store ($M=400$, $S=300$, $k=10$). EC denotes Episode Completeness (gold-turn coverage); AH denotes Answer Hit. O3 bulk-load is on for all rows. Boldface: best AH and EC within each corpus. Graded nDCG for the $M \approx 3000$ grids appears in Table 6.

Corpus	System	EC@10	AH@10
DISC-Law	IVFPQ baseline	0.770	0.540
	+ O1 FlatIP	0.782	0.567
	+ O1+O2 Soft O2	0.888	0.780
Lawyer-LLaMA	IVFPQ baseline	0.698	0.397
	+ O1 FlatIP	0.698	0.397
	+ O1+O2 Soft O2	0.840	0.680

383 5.2. Failure taxonomy

384 We classify each query into four mutually exclusive modes: *complete*
 385 (question and answer in top- k), *incomplete* (question hit / answer miss),
 386 *answer-only* (answer hit / question miss), and *session miss* (no gold-session
 387 turn in top- k). Table 4 reports these rates for FlatIP, Soft O2, hard hy-
 388 dration, and shuffled *sid* on primary non-exact channels under the unified
 389 FlatIP rebuild. Soft O2 converts many incomplete FlatIP cases into com-
 390 plete retrieval while session-miss stays nearly fixed; shuffled *sid* restores high
 391 incompleteness. Hard hydration can drive incompleteness near zero by score
 392 copy and often leads Soft O2 on complete rate (Section 5.4).

393 5.3. Turn-level Soft O2 controls

394 Primary Soft O2 claims use a shared turn-level index. Table 5 reports
 395 Answer Hit for FlatIP, Soft O2, hard hydration, and shuffled *sid* on the pri-
 396 mary Soft O2 transfer channels. Soft O2 beats FlatIP on every non-exact

Table 4: Failure taxonomy at $k=10$ (values in %; rows sum to 100 within rounding). DISC/Lawyer denote LegalEp corpora. Classes: complete (Q+A), incomplete (Q hit / A miss), answer-only, and session miss. Boldface: best complete / lowest incomplete among FlatIP, Soft O2, and shuffled *sid*; hard hydration is unrestricted score copy.

Channel	System	Comp.	Inc.	A-only	Miss
DISC / u-para	FlatIP	68.8	28.8	0.8	1.6
DISC / u-para	Soft O2	88.1	9.9	0.2	1.8
DISC / u-para	Hard	97.2	0.0	0.0	2.8
DISC / u-para	Shuf. O2	61.6	35.8	0.4	2.2
DISC / advice	FlatIP	47.2	17.6	0.0	35.2
DISC / advice	Soft O2	61.0	3.6	0.0	35.4
DISC / advice	Hard	64.8	0.0	0.0	35.2
DISC / advice	Shuf. O2	43.4	21.2	0.0	35.4
Lawyer / u-para	FlatIP	66.9	29.5	2.8	0.8
Lawyer / u-para	Soft O2	94.6	2.8	1.0	1.6
Lawyer / u-para	Hard	97.8	0.0	0.0	2.2
Lawyer / u-para	Shuf. O2	64.9	30.7	2.6	1.8
CAIL / Uk	FlatIP	43.2	56.7	0.0	0.2
CAIL / Uk	Soft O2	93.2	6.7	0.0	0.2
CAIL / Uk	Hard	99.7	0.2	0.0	0.2
CAIL / Uk	Shuf. O2	31.3	68.5	0.0	0.2

channel; shuffled *sid* collapses toward or below FlatIP. Hard hydration—full sibling score copy—can exceed Soft O2 AH on the same rebuild (Table 5); Soft O2 injects siblings at attenuated scores βs with $\beta < 1$ (Tables A.5–A.7). BM25-turn is a same-granularity lexical control (Table A.8). Joint episode / document-level indexes change evidence granularity and are scoped as alternative designs rather than Soft O2 turn-binding baselines (Section 6.3).

5.4. Soft O2 on session (CAIL and LegalEp)

Figure 3 and Table 5 summarize AH@10 on authentic CAIL multi-turn channels at $M \approx 3000$ under O1 FlatIP. Soft O2 raises Answer Hit over FlatIP on every channel (U1 0.948 vs 0.803; Uk 0.932 vs 0.432; U-last 0.957 vs 0.393). Shuffled *sid* removes the gain, tying the improvement to correct episode membership. Hard hydration can raise AH further by copying sibling scores without attenuation (e.g., Uk 0.997 vs Soft O2 0.932). Table 6 preserves the Soft O2>FlatIP ordering under graded nDCG with gold-session

Table 5: Primary Soft O2 controls (AH@10, $M \approx 3000$, unified FlatIP rebuild). Boldface: best AH within each row among FlatIP / Soft O2 / Hard / Shuffled. Soft O2 is compared primarily to FlatIP; hard hydration is unrestricted score copy; shuffled *sid* tests membership dependence.

Corpus / channel	FlatIP	Soft O2	Hard	Shuffled
CAIL / U1	0.803	0.948	0.995	0.660
CAIL / Uk	0.432	0.932	0.997	0.313
CAIL / U-last	0.393	0.957	0.998	0.255
LegalEp-DISC / u-para	0.696	0.883	0.972	0.620
LegalEp-DISC / advice	0.472	0.610	0.648	0.434
LegalEp-Lawyer / u-para	0.697	0.956	0.978	0.675
LegalEp-Lawyer / advice	0.416	0.654	0.674	0.396

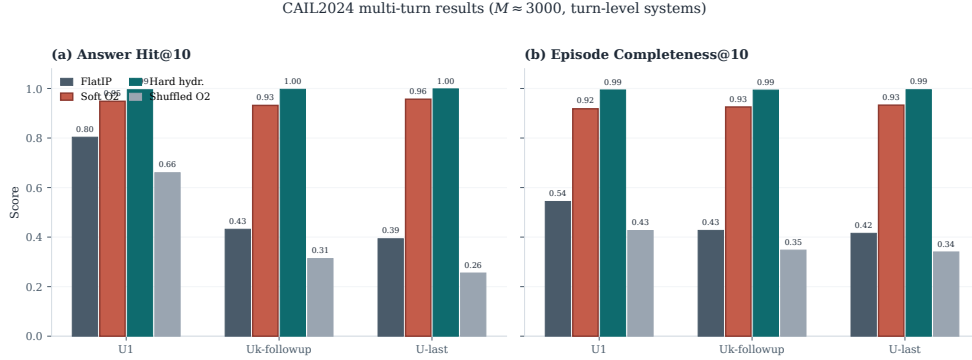


Figure 3: CAIL2024 multi-turn results at tier $M \approx 3000$. Soft O2 improves AH and EC over FlatIP; shuffled *sid* removes the gain; hard hydration can further raise AH/EC.

411 IDCG, and incompleteness falls sharply (Table 4). Full numeric grids appear
 412 in Tables A.5–A.7.

413 Figure 4 reports LegalEp transfer on the same AH grids. Soft O2 also
 414 raises exact replay (DISC 0.916 vs FlatIP 0.714; Lawyer 0.976 vs 0.678),
 415 which we treat as diagnostic relative to paraphrase and advice-recall. Para-
 416 phrase gains are large: 0.883 vs 0.696 on DISC and 0.956 vs 0.697 on
 417 Lawyer. Advice-recall rises to 0.610 from FlatIP 0.472 on DISC and to
 418 0.654 from 0.416 on Lawyer. Hard hydration can still edge Soft O2 on these
 419 channels; shuffled *sid* collapses toward FlatIP. Graded nDCG mirrors the
 420 Soft O2>FlatIP ordering on paraphrase and advice-recall (Table 6). Detailed
 421 grids are in Tables A.6–A.7.

Table 6: Graded nDCG@10 with gold-session IDCG (answer $rel=1.0$, other same-session turns $rel=0.5$) and incompleteness rates on primary non-exact channels. Boldface: Soft O2 better than FlatIP within each channel on nDCG. Soft O2 also exceeds shuffled *sid*.

Corpus	Channel	System	nDCG@10	Incomplete
CAIL	Uk	FlatIP	0.428	56.7%
CAIL	Uk	Soft O2	0.796	6.7%
CAIL	Uk	Shuffled O2	0.349	68.5%
CAIL	U-last	FlatIP	0.400	60.5%
CAIL	U-last	Soft O2	0.796	4.2%
CAIL	U-last	Shuffled O2	0.326	74.3%
LegalEp-DISC	u-para	FlatIP	0.659	28.8%
LegalEp-DISC	u-para	Soft O2	0.742	9.9%
LegalEp-DISC	u-para	Shuffled O2	0.590	35.8%
LegalEp-DISC	advice	FlatIP	0.454	17.6%
LegalEp-DISC	advice	Soft O2	0.518	3.6%
LegalEp-DISC	advice	Shuffled O2	0.411	21.2%
LegalEp-Lawyer	u-para	FlatIP	0.708	29.5%
LegalEp-Lawyer	u-para	Soft O2	0.832	2.8%
LegalEp-Lawyer	u-para	Shuffled O2	0.656	30.7%
LegalEp-Lawyer	advice	FlatIP	0.457	26.2%
LegalEp-Lawyer	advice	Soft O2	0.569	2.0%
LegalEp-Lawyer	advice	Shuffled O2	0.424	27.8%

422 Cross-encoder reranking is a strong neural control at the same turn-level
423 evidence unit: FlatIP+CE reranks the top-30 FlatIP candidates with **bge-**
424 **reranker-v2-m3**. Table 7 reports a secondary CE campaign (separate from
425 the unified Soft O2 rebuild): within that campaign FlatIP+CE is strongest
426 on LegalEp exact channels, while Soft O2 leads on CAIL multi-turn chan-
427 nels (Uk 0.698 vs CE 0.375; U-last 0.762 vs 0.342). A structural limit of
428 the reranker explains the split: CE can refine ranking among candidates
429 already retrieved by FlatIP, but it cannot promote an answer sibling that
430 never entered the top-30 pool. Soft O2 recovers such siblings by propagat-
431 ing attenuated scores through session links, including cases where the stored
432 advice turn has weak direct lexical overlap with the query (paraphrase and
433 advice-recall; Tables 5 and 6).

434 5.5. Soft O2-C and gated Hybrid under Mix

435 When gold answers already share *sid* with the query, Soft O2 is sufficient
436 and cluster binding is unnecessary. Table 8 asks the off-session question under

LegalEp ($M=3000$): Soft O2 gains are strongest on paraphrase and remain positive on advice-recall

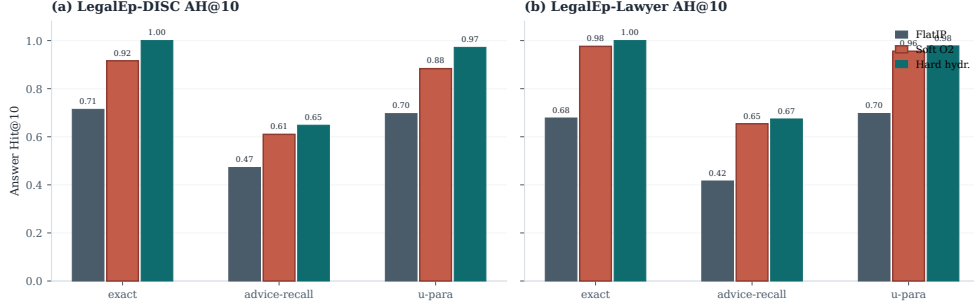


Figure 4: LegalEp AH@10 ($M=3000$, 500 needles). Soft O2 gains are strongest on paraphrase and remain positive on advice-recall; exact transfer is corpus-dependent and treated as diagnostic.

Table 7: Secondary cross-encoder control (AH@10; CE campaign store). FlatIP+CE reranks the top-30 FlatIP candidates with a multilingual cross-encoder. Soft O2 column is from the same CE campaign for within-table comparison; primary Soft O2 vs FlatIP AH uses the unified rebuild (Tables 5–A.7). Boldface: best AH within each row.

Corpus / channel	FlatIP	FlatIP+CE	Soft O2
CAIL / U1	0.585	0.688	0.788
CAIL / Uk	0.265	0.375	0.698
CAIL / U-last	0.228	0.342	0.762
LegalEp-DISC / exact	0.572	0.722	0.572
LegalEp-Lawyer / exact	0.636	0.708	0.636

437 the Mix protocol of Section 4.3: 70% cross-session Sa/Sb gold and 30% intact
 438 same-session gold, with FlatIP, Soft O2, Soft O2-C, and gated Hybrid sharing
 439 unrestricted dense retrieval. Hybrid uses a direct-dense gate: cluster binding
 440 fires only on turn ids recorded *before* Soft O2 session expansion.

441 On LegalEp Mix stores, gated Hybrid is the strongest overall system
 442 (DISC AH 0.542, Lawyer 0.536; McNemar mid- $p=0.0005^{**}$ / 0.0022^{**} vs
 443 Soft O2) and also leads AH_{cross} . Ungated Soft O2-C raises DISC AH_{cross}
 444 to 0.506 from Soft O2 0.460 and Lawyer cross to 0.480 from 0.440, but re-
 445 mains weaker than Hybrid once session binding is composed under the gate.
 446 On LegalMem-MT Mix the picture reverses: Soft O2 retains the best overall
 447 AH (0.602), while Soft O2-C (0.440) and Hybrid (0.562) trail; FlatIP even
 448 leads the cross stratum (0.520). Cluster binding is therefore conditional on
 449 corpus and cluster purity—helpful as a gated add-on on LegalEp advice-recall

Mix, but not uniformly beneficial on CAIL multi-turn Mix under the same protocol. Same-session Soft O2 vs Soft O2-C contrasts remain Soft O2-led (Appendix Appendix A, Table A.1).

Table 8: Mix evaluation of Soft O2-C and gated Hybrid with the direct-dense cluster trigger (same store, unrestricted dense; AH@10). Gold is 70% cross-session Sa/Sb + 30% intact same-session. Boldface: best overall AH and best cross-session AH within each corpus. Significance vs Soft O2: * $p < 0.05$, ** $p < 0.01$.

Corpus	System	AH	AH _{cross}	AH _{same}	mid- p vs Soft O2
LegalEp-DISC Mix	FlatIP	0.494	0.491	0.500	—
	Soft O2	0.492	0.460	0.567	—
	Soft O2-C	0.498	0.506	0.480	0.7275
	Hybrid	0.542	0.526	0.580	0.0005**
LegalEp-Lawyer Mix	FlatIP	0.460	0.451	0.480	—
	Soft O2	0.492	0.440	0.613	—
	Soft O2-C	0.464	0.480	0.427	0.1284
	Hybrid	0.536	0.520	0.573	0.0022**
LegalMem-MT Mix	FlatIP	0.530	0.520	0.553	—
	Soft O2	0.602	0.489	0.867	—
	Soft O2-C	0.440	0.460	0.393	0.0000**
	Hybrid	0.562	0.446	0.833	0.0594

Hybrid uses the direct-dense gate. Soft O2-C mid- p vs Soft O2 may favor either side; on LegalMem-MT the ** marks Soft O2-C significantly *below* Soft O2. Cluster co-membership of Sa/Sb pairs follows Section 4.3.

5.6. Scale behaviour

Figure 5 and Table A.3 report quality and warm-query latency on the LegalEp DISC exact channel ($Q=100$; three repeats; archive sizes 100, 400, 1600, and 6400). Soft O2 retains an AH advantage at each tier. IVFPQ collapses at $M=1600$ (AH 0.330), which motivates O1, while absolute latency grows with M . We treat O1 FlatIP as an engineering prerequisite for Soft O2 at these scales and do not claim deployment readiness beyond the measured curve.

End-to-end QA audit (summary).. A dual-protocol QA audit ($N=270$ per corpus; generator **qwen3:14b**, independent judge **qwen3:32b**) reports judged answerability under Soft O2 retrieval alone; it is not a paired FlatIP vs

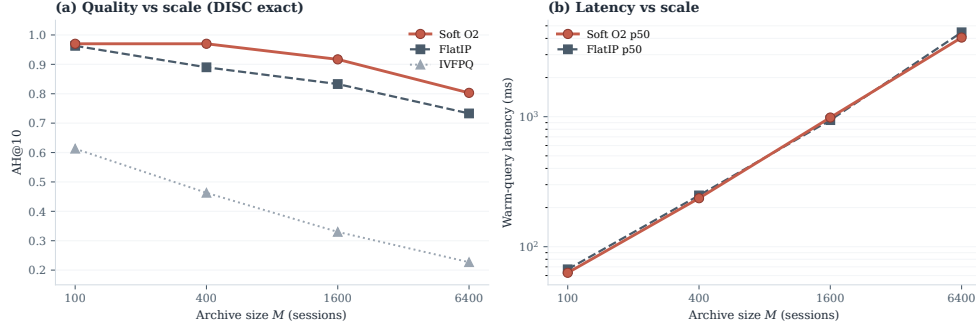


Figure 5: Scale curve on DISC-Law exact. Soft O2 keeps an AH edge over FlatIP as M grows; warm-query latency scales with archive size; IVFPQ quality collapses at moderate M .

467 Soft O2 generation experiment. Full protocol and stratified results appear in
 468 Section 6.2. On a smaller shared store ($M=400$, five seeds), Soft O2 AH on
 469 LegalEp-DISC is 0.851 ± 0.008 versus FlatIP 0.698 ± 0.040 .

470 6. Discussion

471 6.1. Findings and scope

472 Episode incompleteness is a substantial FlatIP failure under same-domain
 473 interference; Soft O2 converts many incomplete retrievals into complete ones
 474 without shrinking session-miss rates (Table 4). O1 restores score fidelity;
 475 Soft O2 supplies the main gain through soft sibling inheritance; O3 makes
 476 large shared stores tractable. The gap-ratio band $\rho < \beta < 1$ and the AH
 477 sensitivity plateau $\{0.9, 0.95, 0.98\}$ jointly motivate freezing $\beta=0.98$ below
 478 hard hydration. Primary Soft O2 vs FlatIP gains on CAIL and LegalEp
 479 remain Holm-significant (Table A.2); paraphrase and advice-recall are the
 480 main transfer channels, with graded nDCG agreeing (Tables 5–6). Under
 481 Mix, gated Hybrid improves overall AH over Soft O2 on LegalEp-DISC and
 482 LegalEp-Lawyer, while Soft O2 remains strongest on LegalMem-MT and
 483 on intact same-session gold (Tables 8–A.1). Turn-level BM25, RRF, and
 484 CE place Soft O2 among standard IR alternatives Robertson and Zaragoza
 485 (2009); Cormack et al. (2009); Nogueira et al. (2019); joint-episode indexes
 486 remain a coarser design outside this protocol.

487 6.2. RAG application perspective

488 Consultation-memory retrieval is judged downstream by whether an LLM
 489 can answer from recovered evidence rather than confabulate. We audit this
 490 end-to-end with a dual-protocol pipeline on Soft O2 retrieval only: genera-
 491 tor `qwen3:14b` and independent judge `qwen3:32b` are distinct models. Per
 492 corpus we pool $N=270$ judged instances as P1 $n=120$ plus P2 $n=150$. The
 493 independent judge assigns a continuous evidence-answerability score in $[0, 1]$
 494 from the retrieved context and the generated reply; Evidence-Answerability
 495 Rate (EAR) is the fraction of instances with judge score ≥ 0.7 . Table 9
 496 reports pooled EAR and the mean judge score on a stratified subsample
 497 ($n=90$ /corpus; 30 each of exact, paraphrase, and follow-up), labelled “Strati-
 498 fied judge” in the table. In the complete-episode stratum—both question
 499 and answer turns in top- k —judged answerability is higher in this single-
 500 system audit (pooled EAR 0.867 on DISC and 0.859 on Lawyer; Wilson 95%
 501 CI $[0.83, 0.90]$ / $[0.82, 0.89]$). Follow-up queries remain the hardest stratum
 502 (independent-judge EAR 0.517 / 0.480), consistent with episodic incomplete-
 503 ness persisting even when surface-form overlap is low.

504 A blind two-annotator legal audit ($n=60$ /corpus) separates retrieval fail-
 505 ures (gold episode absent from top- k) from generation failures given retrieved
 506 evidence. Annotators used a binary answerability label on each instance;
 507 agreement is Cohen’s κ computed on those paired labels (0.71 on DISC, 0.68
 508 on Lawyer). Retrieval-failure share is 0.283 / 0.317. Wrong-session con-
 509 tamination rates stay low (0.083 / 0.100), suggesting that complete episode
 510 retrieval under Soft O2 is associated with lower rates of a common RAG
 511 failure mode in which the model answers from a neighbouring client’s matter.
 512 Hallucinated-legal-basis rates (0.117 / 0.133) remain non-zero: grounding in
 513 retrieved episodes does not guarantee statutory correctness. AH and EAR
 514 measure evidence availability and judged answerability respectively; statu-
 515 tory correctness, citation validity, and professional liability lie outside this
 516 evaluation.

517 6.3. Limitations

518 LegalEp episodes are consultation pairs; multi-turn authenticity remains
 519 reserved for CAIL.

520 *Dependency on session segmentation and cluster quality..* Soft O2 relies on
 521 correct session identifiers (*sid*). In uncured client logs, boundary noise—
 522 mid-session topic shifts, multi-intent interruptions, or automated segmenta-

Table 9: End-to-end QA audit ($N=270$ per corpus; generator `qwen3:14b`, independent judge `qwen3:32b`). Pooled EAR: fraction with judge score ≥ 0.7 . Stratified judge: mean judge score on $n=90$ /corpus (exact/paraphrase/follow-up). Human κ : Cohen’s κ on binary answerability labels from two blind annotators ($n=60$ /corpus).

Corpus	Pooled EAR	Stratified judge	Human κ
DISC-Law	0.867	0.757	0.71
Lawyer-LLaMA	0.859	0.729	0.68

tion drift—can corrupt sibling sets. If unrelated turns share a *sid*, attenuated inheritance may propagate false context; if a true advice turn is split into another session, Soft O2 cannot bind it. Soft O2-C likewise depends on KMeans cluster purity. Under the Mix protocol, Sa/Sb pairs receive a shared cluster id so binding can be tested when thematic co-membership is present (Section 4.3); in fully unsupervised consolidation, noisier boundaries shrink the margin over dense retrieval. Deployments on raw consultation streams therefore need reliable upstream session segmentation, and Soft O2-C should be treated as gated rather than always-on.

Joint-episode indexing.. Concatenating full question-answer pairs into episode documents prevents incompleteness by changing the evidence unit from turns to merged documents (Section 2). That design lies outside the Soft O2 turn-binding protocol. A matched comparison under online updates, latency, and fine-grained advice isolation is left to future work.

Embedding model and other scope limits.. All main grids use `nomic-embed-text-v2-moe`. A Chinese-specialized encoder such as `bge-large-zh-v1.5` could alter absolute dense ranks and the measured ρ distribution, and might shrink or widen Soft O2’s relative gain wherever the bi-encoder already places answer turns inside top- k . We did not re-run the full Soft O2 grids under alternative encoders; the binding operator itself is encoder-agnostic once fused scores are available. The ρ statistics use an exact-channel query proxy and the strongest measured distractor; paraphrase and advice-recall gap laws may shift the feasible band slightly, though the AH sweep keeps a stable high- β region on those channels. Main $M \approx 3000$ tables use a fixed seed; five-seed variance is reported only for the smaller $M=400$ store. LLM paraphrases need human audit. Statutory correctness, citation validity, and professional liability remain unevaluated. Because many channels are tested, Holm-adjusted

550 p -values accompany the Soft O2 vs FlatIP family (Table A.2); Mix Hybrid sig-
 551 nificance is reported only vs Soft O2 within Table 8 and is corpus-conditional.

552 *Latency and deployment scope.* Scale curves cover archive sizes 100, 400,
 553 1600, and 6400 under a fixed hardware stack (Table A.3). At $M=6400$, warm-
 554 query p50 latency exceeds 4s (4061ms for Soft O2; 4458ms for FlatIP)—
 555 unacceptable for millisecond interactive search. BIMS-LEGAL is there-
 556 fore best positioned for high-precision offline or near-real-time consultation-
 557 memory recovery—for example, pre-meeting case review, audit replay, or
 558 batch RAG over a firm’s advice archive—rather than as a sub-millisecond
 559 search engine. Future work will combine approximate nearest-neighbour in-
 560 dexes (e.g., HNSW) with Soft O2 and Soft O2-C binding layered on top
 561 of faithful candidate retrieval, preserving episode completion while reducing
 562 query latency. Behavior at roughly 10^5 sessions is not measured, and we do
 563 not claim asymptotic scalability from O3 wall-clock alone.

564 7. Conclusion

565 BIMS-LEGAL recovers prior advice from Chinese legal consultation
 566 archives under same-domain interference with a dual store: an episodic turn
 567 index and a KMeans semantic store. Soft O2 completes missing siblings
 568 by attenuated session binding ($\beta=0.98$), selected inside the gap-ratio band
 569 via an Answer Hit sensitivity sweep (Table A.4); hard hydration remains a
 570 stronger unattenuated upper bound. Soft O2-C extends the same operator
 571 to thematic clusters under Mix reconstructions as a conditional mechanism
 572 test, with O1 FlatIP and O3 bulk-load as the shared-store implementation
 573 operators. On CAIL multi-turn and LegalEp paraphrase /advice-recall chan-
 574 nels, Soft O2 improves Answer Hit and graded nDCG over FlatIP and cuts
 575 incompleteness; shuffled-*sid* collapses the Soft O2 gain. Under Mix, gated Hy-
 576 brid recovers additional LegalEp gold when cluster co-membership is present,
 577 while Soft O2 remains preferable on LegalMem-MT Mix and on intact same-
 578 session gold. A Soft O2 QA audit links complete-episode retrieval to judged
 579 answerability with low wrong-session contamination; it is not a paired FlatIP
 580 vs Soft O2 generation comparison.

581 **Declarations**

582 *Funding*

583 Computational resources were provided by the Data Law Laboratory,
584 China University of Political Science and Law.

585 *Conflict of interest*

586 The authors declare no competing interests.

587 *Ethics approval*

588 This study uses publicly released research corpora and excludes confidential
589 client data. The system is a research prototype for consultation-memory
590 retrieval.

591 *Data availability*

592 Code, processed evaluation corpora, and primary Soft O2 metric artifacts
593 (including paired per-query Answer Hit used for Holm tests) are available at
594 <https://github.com/Kilimajaro/soft-episode-binding-legal-memory>
595 (archived release to accompany camera-ready). Upstream corpora remain
596 under their original licenses: CAIL2024 consultation tracks; Hugging Face
597 ShengbinYue/DISC-Law-SFT and Skepsun/lawyer_llama_data. Processed
598 LegalEp /LegalMem artifacts are derived research releases; redistributors
599 should consult each upstream licence before reuse.

600 *Code availability*

601 The BIMS-LEGAL implementation (O1 FlatIP, Soft O2, O3 bulk-load,
602 Soft O2-C), corpus builders, Mix constructors, and the scripts that regenerate
603 Soft O2 tables, figures, and Holm tests from one metric family ship in the
604 same repository.

605 *Author contributions*

606 L.X. (University of Winnipeg) designed BIMS-LEGAL and the
607 Soft O2 /Soft O2-C evaluation, implemented the system, ran the experi-
608 ments, and wrote the manuscript. L.H. (corresponding author; CUPL Data
609 Law Lab and Institute for Data Law) provided methodology guidance, revision,
610 and supervision.

611 **References**

- 612 Ashley, K.D., 2017. Artificial Intelligence and Legal Analytics: New Tools
613 for Law Practice in the Digital Age. Cambridge University Press.
- 614 Askari, A., Aliannejadi, M., Abolghasemi, A., Kanoulas, E., Verberne, S.,
615 2024. Answer retrieval in legal community question answering, in: Ad-
616 vances in Information Retrieval (ECIR), Springer. pp. 477–485.
- 617 Atkinson, R.C., Shiffrin, R.M., 1971. The control of short-term memory.
618 Scientific American 225, 82–91.
- 619 Callan, J.P., Lu, Z., Xu, S., 1994. Passage-level evidence in document re-
620 trieval, in: Proceedings of the 17th Annual International ACM SIGIR
621 Conference on Research and Development in Information Retrieval, pp.
622 302–310.
- 623 Chen, J., Xiao, S., Zhang, P., Luo, K., Lian, D., Liu, Z., 2024. M3-
624 embedding: Multi-linguality, multi-functionality, multi-granularity text
625 embeddings through self-knowledge distillation, in: Findings of the associ-
626 ation for computational linguistics: ACL 2024, pp. 2318–2335.
- 627 Cormack, G.V., Clarke, C.L.A., Buettcher, S., 2009. Reciprocal rank fusion
628 outperforms condorcet and individual rank learning methods, in: Proceed-
629 ings of the 32nd International ACM SIGIR Conference on Research and
630 Development in Information Retrieval, pp. 758–759.
- 631 Dai, Z., Xiong, C., Liu, Z., Callan, J., 2019. Context-aware sentence/passage
632 encoding for answering factoid questions, in: Proceedings of the 2019 Con-
633 ference on Empirical Methods in Natural Language Processing and the
634 9th International Joint Conference on Natural Language Processing, pp.
635 241–251.
- 636 Gutiérrez, B.J., Shu, Y., Gu, Y., Yasunaga, M., Su, Y., 2024. Hipporag:
637 Neurobiologically inspired long-term memory for large language models.
638 Advances in neural information processing systems 37, 59532–59569.
- 639 Huang, Q., Tao, M., Zhang, C., An, Z., Jiang, C., Chen, Z., Wu, Z., Feng,
640 Y., 2023. Lawyer llama technical report. [arXiv:2305.15062](https://arxiv.org/abs/2305.15062).
- 641 Johnson, J., Douze, M., Jégou, H., 2019. Billion-scale similarity search with
642 GPUs. IEEE Transactions on Big Data 7, 535–547.

- 643 Karpukhin, V., Oguz, B., Min, S., Lewis, P., Wu, L., Edunov, S., Chen,
644 D., Yih, W.t., 2020. Dense passage retrieval for open-domain question
645 answering, in: Proceedings of the 2020 Conference on Empirical Methods
646 in Natural Language Processing (EMNLP), pp. 6769–6781.
- 647 Kumaran, D., Hassabis, D., McClelland, J.L., 2016. What learning systems
648 do intelligent agents need? complementary learning systems theory up-
649 dated. Trends in cognitive sciences 20, 512–534.
- 650 Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küt-
651 tler, H., Lewis, M., Yih, W.t., Rocktäschel, T., et al., 2020. Retrieval-
652 augmented generation for knowledge-intensive nlp tasks. Advances in neural
653 information processing systems 33, 9459–9474.
- 654 Liu, B., Wu, Y., Zhang, F., Liu, Y., Wang, Z., Li, C., Zhang, M., Ma, S., 2022.
655 Query generation and buffer mechanism: Towards a better conversational
656 agent for legal case retrieval. Information Processing & Management 59,
657 103051.
- 658 Louis, A., van Dijck, G., Spanakis, G., 2024. Interpretable long-form legal
659 question answering with retrieval-augmented large language models, in:
660 Proceedings of the AAAI Conference on Artificial Intelligence, pp. 22266–
661 22275.
- 662 Maharana, A., Lee, D.H., Tulyakov, S., Bansal, M., Barbieri, F., Fang, Y.,
663 2024. Evaluating very long-term conversational memory of llm agents, in:
664 Proceedings of the 62nd Annual Meeting of the Association for Computa-
665 tional Linguistics (Volume 1: Long Papers), pp. 13851–13870.
- 666 Martinez-Gil, J., 2023. A survey on legal question–answering systems. Com-
667 puter Science Review 48, 100552.
- 668 McClelland, J.L., McNaughton, B.L., O’Reilly, R.C., 1995. Why there are
669 complementary learning systems in the hippocampus and neocortex: in-
670 sights from the successes and failures of connectionist models of learning
671 and memory. Psychological review 102, 419.
- 672 Mo, F., Gao, Y., Wu, Z., Liu, X., Chen, P., Li, Z., Wang, Z., Li, X., Jiang,
673 M., Nie, J.Y., 2026. Leveraging historical information to boost retrieval-
674 augmented generation in conversations. Information Processing & Man-
675 agement 63, 104449.

- 676 Modarressi, A., Imani, A., Fayyaz, M., Schütze, H., 2023. Ret-llm: Towards
677 a general read-write memory for large language models. arXiv preprint
678 arXiv:2305.14322 .
- 679 Nogueira, R., Yang, W., Cho, K., Lin, J., 2019. Multi-stage document rank-
680 ing with BERT. arXiv:1910.14497.
- 681 Nussbaum, Z., et al., 2025. Nomic embed multimodal: Interleaved text, im-
682 age, and datasets for general-purpose vision-language embeddings. Model
683 card: **nomic-embed-text-v2-moe** (Nomic AI). Embedding model used for
684 turn encoding in this work.
- 685 Packer, C., Wooders, S., Lin, K., Fang, V., Patil, S.G., Stoica, I., Gonzalez,
686 J.E., 2023. Memgpt: Towards llms as operating systems. arXiv preprint
687 arXiv:2310.08560 .
- 688 Rasmussen, P., Paliychuk, P., Beauvais, T., Ryan, J., Chalef, D., 2025. Zep:
689 A temporal knowledge graph architecture for agent memory. arXiv preprint
690 arXiv:2501.13956 .
- 691 Robertson, S., Zaragoza, H., 2009. The probabilistic relevance framework:
692 BM25 and beyond. Foundations and Trends in Information Retrieval 3,
693 333–389.
- 694 Tan, H., Zhang, Z., Ma, C., Chen, X., Dai, Q., Dong, Z., 2025. Mem-
695 bench: Towards more comprehensive evaluation on the memory of llm-
696 based agents, in: Findings of the Association for Computational Linguis-
697 tics: ACL 2025, pp. 19336–19352.
- 698 Tulving, E., et al., 1972. Episodic and semantic memory. Organization of
699 memory 1, 1.
- 700 Wang, Y., Gao, Y., Chen, X., Jiang, H., Li, S., Yang, J., Yin, Q., Li, Z.,
701 Li, X., Yin, B., et al., 2024. Memoryllm: Towards self-updatable large
702 language models. arXiv preprint arXiv:2402.04624 .
- 703 Wu, D., Wang, H., Yu, W., Zhang, Y., Chang, K.W., Yu, D., 2024. Long-
704 memeval: Benchmarking chat assistants on long-term interactive memory.
705 arXiv preprint arXiv:2410.10813 .

- 706 Wu, Y., Liang, S., Zhang, C., Wang, Y., Zhang, Y., Guo, H., Tang, R.,
 707 Liu, Y., 2025. From human memory to ai memory: A survey on memory
 708 mechanisms in the era of llms. [arXiv:2504.15965](#).
- 709 Yue, S., Chen, W., Wang, S., Li, B., Shen, C., Liu, S., Zhou, Y., Xiao, Y.,
 710 Yun, S., Huang, X., Wei, Z., 2023. Disc-lawllm: Fine-tuning large language
 711 models for intelligent legal services. [arXiv:2309.11325](#).
- 712 Yue, S., Liu, S., Zhou, Y., Shen, C., Wang, S., Xiao, Y., Li, B., Yun, S.,
 713 Shen, X., Chen, W., et al., 2024. Lawllm: Intelligent legal system with
 714 legal reasoning and verifiable retrieval, in: International Conference on
 715 Database Systems for Advanced Applications, Springer. pp. 304–321.
- 716 Zaheer, M., Kottur, S., Ravanbakhsh, S., Poczos, B., Salakhutdinov, R.,
 717 Smola, A.J., 2017. Deep sets, in: Advances in Neural Information Process-
 718 ing Systems.
- 719 Zhang, Y., Zhai, S., Meng, Y., Bi, S., Chen, Y., Qi, G., 2024. Event is
 720 more valuable than you think: Improving the similar legal case retrieval
 721 via event knowledge. *Information Processing & Management* 61, 103729.
- 722 Zhong, W., Guo, L., Gao, Q., Ye, H., Wang, Y., 2024. Memorybank: En-
 723 hancing large language models with long-term memory, in: Proceedings of
 724 the AAAI Conference on Artificial Intelligence, pp. 19724–19731.

725 **Appendix A. Numeric grids and controls**

726 Same-session Soft O2-C contrasts, significance tests, scale curves, and full
 727 Soft O2 numeric grids appear below.

Table A.1: Same-session Soft O2 vs ungated Soft O2-C (AH@10). Soft O2 is preferable when gold already shares *sid* with the query; ungated Soft O2-C can surface thematic confusers. Boldface: better AH within each row. Mix Soft O2-C results appear in Table 8.

Corpus / channel	Soft O2	Soft O2-C
LegalEp-DISC / exact	0.836	0.744
LegalEp-DISC / u-para	0.807	0.730
LegalEp-DISC / advice	0.548	0.490
LegalEp-Lawyer / exact	0.818	0.710
LegalEp-Lawyer / u-para	0.817	0.733
LegalEp-Lawyer / advice	0.546	0.448
LegalMem-MT / U1	0.903	0.712
LegalMem-MT / Uk	0.793	0.385

Table A.2: Primary Soft O2 vs FlatIP family: McNemar mid- p on paired Answer Hit from the unified FlatIP rebuild, with Holm step-down adjustment ($m=9$). Each ΔAH matches Soft O2–FlatIP in the main Soft O2 tables; paired outcomes are released with the paper artifacts.

Corpus / channel	Contrast	ΔAH	mid- p	Holm p
CAIL / Uk	O2 vs FlatIP	+0.500	3.69e-69	2.95e-68
CAIL / U1	O2 vs FlatIP	+0.145	1.15e-16	3.44e-16
CAIL / U-last	O2 vs FlatIP	+0.563	3.19e-85	2.87e-84
LegalEp-DISC / exact	O2 vs FlatIP	+0.202	2.32e-19	9.28e-19
LegalEp-Lawyer / exact	O2 vs FlatIP	+0.298	1.16e-39	8.12e-39
LegalEp-DISC / u-para	O2 vs FlatIP	+0.187	1.28e-16	3.44e-16
LegalEp-Lawyer / u-para	O2 vs FlatIP	+0.259	2.07e-32	1.04e-31
LegalEp-DISC / advice	O2 vs FlatIP	+0.138	1.52e-15	1.52e-15
LegalEp-Lawyer / advice	O2 vs FlatIP	+0.238	7.84e-33	4.70e-32

Table A.3: Scale curve on DISC-Law (mean over 3 repeats). Latency is warm-query p50/p95. Boldface: best AH@10 within each M block; lowest p50/p95 within each M block.

M	Mode	AH@10	Build (s)	p50 (ms)	p95 (ms)
100	FlatIP	0.963	3.0	67	80
100	IVFPQ	0.613	25.2	40	54
100	Soft O2	0.970	3.2	63	69
400	FlatIP	0.890	26.5	248	305
400	IVFPQ	0.463	89.5	171	204
400	Soft O2	0.970	30.9	236	313
1600	FlatIP	0.833	359.8	941	1278
1600	IVFPQ	0.330	251.4	669	751
1600	Soft O2	0.917	393.7	985	1172
6400	FlatIP	0.733	4722.8	4458	4969
6400	IVFPQ	0.227	1072.1	2182	3021
6400	Soft O2	0.803	5245.7	4061	4503

Table A.4: Soft O2 AH@10 sensitivity sweep over $\beta \in \{0.5, 0.7, 0.9, 0.95, 0.98, 1.0\}$ on the CAIL Uk-followup and LegalEp exact/paraphrase evaluation channels (LegalMem-MT β -campaign stores; not held-out). Used with the gap-ratio analysis to freeze default $\beta=0.98$. Peak AH lies in $\{0.9, 0.95, 0.98\}$; $\beta=0.5$ weakens binding. Not a primary Soft O2 result table; absolute AH values are not comparable to the unified FlatIP-rebuild Tables 5–A.7.

Corpus	0.5	0.7	0.9	0.95	0.98	1.0
CAIL / Uk	0.323	0.537	0.697	0.700	0.698	0.662
LegalEp-Lawyer / exact	0.608	0.654	0.754	0.754	0.752	0.752
LegalEp-Lawyer / u-para	0.631	0.637	0.717	0.745	0.759	0.749
LegalEp-DISC / u-para	0.551	0.551	0.618	0.648	0.648	0.642

Table A.5: CAIL2024 shared-corpus results ($M \approx 3000$). Boldface marks the best AH@10 and best EC@10 within each channel (ties bolded jointly). **: Soft O2 vs FlatIP, McNemar mid- $p < 0.01$. nDCG@10 uses gold-session IDCg.

Channel	System	AH@10	EC@10	nDCG@10
U1	FlatIP	0.803	0.544	0.538
	Soft O2	0.948	0.918	0.767
	Hard hydr.	0.995	0.994	0.859
	Shuffled O2	0.660	0.427	0.393
Uk-followup	FlatIP	0.432	0.427	0.428
	Soft O2	0.932	0.925	0.796
	Hard hydr.	0.997	0.994	0.853
	Shuffled O2	0.313	0.348	0.349
U-last	FlatIP	0.393	0.415	0.400
	Soft O2	0.957	0.933	0.796
	Hard hydr.	0.998	0.995	0.861
	Shuffled O2	0.255	0.340	0.326

728 Soft O2 improves AH/EC over FlatIP; hard hydration uses unrestricted score copy
729 and can lead both.

Table A.6: LegalEp-DISC ($M=3000$, 500 needles). Boldface: best AH@10 and best EC@10 within each channel. * $p < 0.05$, ** $p < 0.01$ (Soft O2 vs FlatIP, McNemar mid- p). Hard hydration coincides with session-max aggregation; only hard hydration is shown. nDCG@10 uses gold-session IDCg.

Channel	System	AH@10	EC@10	nDCG@10
exact	FlatIP	0.714	0.857	0.700
	Soft O2	0.916	0.958	0.792
	Hard hydr.	1.000	1.000	0.873
	Shuffled O2	0.642	0.821	0.626
advice-recall	FlatIP	0.472	0.560	0.454
	Soft O2	0.610	0.628	0.518
	Hard hydr.	0.648	0.648	0.563
	Shuffled O2	0.434	0.540	0.411
u-param	FlatIP	0.696	0.836	0.659
	Soft O2	0.883	0.932	0.742
	Hard hydr.	0.972	0.972	0.816
	Shuffled O2	0.620	0.797	0.590

Table A.7: LegalEp-Lawyer ($M=3000$, 500 needles). Boldface: best AH@10 and best EC@10 within each channel. * $p<0.05$, ** $p<0.01$ (Soft O2 vs FlatIP). Hard hydration coincides with session-max aggregation; only hard hydration is shown. nDCG@10 uses gold-session IDCG.

Channel	System	AH@10	EC@10	nDCG@10
exact	FlatIP	0.678	0.839	0.710
	Soft O2	0.976	0.988	0.851
	Hard hydr.	1.000	1.000	0.880
	Shuffled O2	0.654	0.827	0.656
advice-recall	FlatIP	0.416	0.546	0.457
	Soft O2	0.654	0.664	0.569
	Hard hydr.	0.674	0.674	0.591
	Shuffled O2	0.396	0.535	0.424
u-param	FlatIP	0.697	0.830	0.708
	Soft O2	0.956	0.965	0.832
	Hard hydr.	0.978	0.978	0.847
	Shuffled O2	0.675	0.815	0.656

Table A.8: Turn-level BM25 and dense+BM25 RRF (AH@10). Boldface: higher AH within each row among reported turn-level cells. BM25-joint / joint-episode document indexes are omitted: they change evidence granularity and were not run as Soft O2 protocol competitors.

Corpus	Channel	AH@10
LegalEp-DISC	exact (BM25-turn)	0.802
LegalEp-DISC	advice-recall (BM25-turn)	0.382
LegalEp-DISC	exact (dense+BM25 RRF)	0.780
LegalEp-DISC	advice (dense+BM25 RRF)	0.394
CAIL	U1 / Uk / U-last (BM25-turn)	0.748 / 0.482 / 0.413
LegalEp-Lawyer	exact / advice (BM25-turn)	0.784 / 0.460